Brenda Ntinyari Kinoti

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RESEARCH INTERESTS

Human-computer interaction and technology ethics and governance. I'm interested in how automated decision systems in finance, insurance and health get built and deployed in African contexts, and what they mean for the people on the receiving end. I come to these questions as a practitioner who has built such systems in Kenyan banking and insurance.

EDUCATION

Master of Science in Data Analytics in Business, Seattle Pacific University, Seattle, WA. Expected Spring 2027. GPA: 3.98/4.0

Bachelor of Science in Computer Science, University of Eldoret, Kenya. 2018

Data Science Nanodegree, Moringa School, Nairobi, Kenya. 2024

HONORS

- Beta Gamma Sigma, international business honor society. Inducted May 2026

RESEARCH AND WRITING

From Keywords to Context: Domain-Specific Retrieval-Augmented Generation as the Next Evolution of Enterprise Knowledge Management. Research paper, Knowledge Management and Business Intelligence, Seattle Pacific University, 2026. Argues that domain-specific retrieval-augmented generation systems are the next evolution of enterprise knowledge management infrastructure, drawing on my experience building one for a bank.

Detecting Conflict-of-Interest Exposure in Industry Payments to Clinicians: A Batch, Streaming, and Machine Learning Pipeline for Mephisto Health System. Research paper, Big Data and Cloud Platforms, Seattle Pacific University, 2026. Designs and evaluates a pipeline on the CMS Open Payments dataset (about 16 million records) that flags clinicians whose industry payments point to conflict-of-interest exposure.

Beyond Governance: Information Systems Failures and the 2016 Collapse of Chase Bank (Kenya) Limited. Case study, Information Systems Security, Seattle Pacific University, 2026. Analyzes the governance and security failures behind the collapse using the CISSP framework.

SIM Swap Fraud on M-Pesa. Product analysis and demo, Information Systems Security, Seattle Pacific University, 2026. Examines how AI-based anomaly detection (Darktrace) applies to SIM swap fraud against mobile money users in Kenya.

Patient Pharmacovigilance Text Mining. Data mining project, Seattle Pacific University, Feb 2026. Text mining of 65,000 patient drug reviews to test whether review text alone predicts a patient's drug experience, and whether complaint patterns in negative reviews reveal safety signals.

PROFESSIONAL EXPERIENCE

Dispatcher, Office of Safety & Security, Seattle Pacific University. Jun 2026 to present, Seattle, WA

- Answer emergency and non-emergency calls, assess what each needs, and dispatch patrol officers to respond and log calls and officer activity in real time, and write incident reports that feed the department's records.
- Coordinate with police, fire and medical services when an incident needs outside responders.
- Monitor campus surveillance cameras and alarm systems, and flag activity that needs a response.

Data Scientist, Absa Bank Kenya PLC. Nov 2024 to Feb 2025, Nairobi, Kenya

- Designed and built a document retrieval system for semantic search across financial product documentation, using FastAPI, LangChain and a FAISS vector database.
- Engineered the document processing pipeline for the product catalogue PDF, with chunking strategies and sentence embeddings from Hugging Face (all-MiniLM-L12-v2).
- Integrated the Mistral-7B-Instruct LLM through a custom API endpoint to generate context-aware answers with real-time streaming.
- Deployed the application on AWS EC2, connected to an internal sandbox for testing, and documented the API for future teams.

Data Scientist and MLOps, Innovex Solutions. Jan 2024 to Oct 2024, Nairobi, Kenya

- Automated vehicle damage assessment for insurance claims by architecting an end-to-end ML pipeline for an InsurTech platform, reducing manual processing time by 60%.
- Built a computer vision system using YOLO and Mask R-CNN for vehicle part identification and damage classification.
- Developed a CNN multiclass image classifier in TensorFlow and Keras, reaching 75% accuracy across 7 vehicle body types.
- Modeled latent and pre-existing vehicle damage patterns with K-means clustering and association rule mining (Apriori, FP-growth).
- Deployed models on AWS SageMaker and tracked customer acquisition metrics in Tableau and Power BI dashboards.

Junior Data Scientist, Innovex Solutions. Jun 2023 to Dec 2023, Nairobi, Kenya

- Ran exploratory analysis, preprocessing and feature engineering on insurance datasets in Python and SQL.
- Built statistical models of customer behavior and profitability, and presented customer segment findings to business leaders through reports and Tableau dashboards.

Business Intelligence Developer and Analyst, Innovex Solutions. Apr 2021 to May 2023, Nairobi, Kenya

- Managed the online auction platform eauqtion.com, processing over 100 auctions worth more than Ksh. 1 billion ($7M+) in assets, with full financial reconciliation reporting.
- Designed and implemented an e-commerce procurement framework for insurance salvage disposal, used across multiple institutions.
- Led product operations for the InsurTech claim management system FLIP, from feature definition through development to service delivery.
- Built KPI dashboards and financial reports for executive leadership.

TEACHING AND MENTORING

Computer instructor, National Youth Service, Kenya. 2018 to 2021

- Taught and mentored rotating classes of 30 to 40 National Youth Service recruits, most of whom had never used a computer, guiding them from a basic introduction to computers through database management.
- Taught the same courses twice a week to flower farm workers at Bohemian Farm (formerly Oserian) in Naivasha, who had never had computer classes or a standard formal education.

Mentor, women in tech and students, Kenya. 2025 to present

- Mentor early-career women in tech from Innovex and students from my home community in Meru.

Mentee, CAL Mentor Program, Seattle Pacific University. 2025 to present

SELECTED PROJECTS

More on GitHub.

Content Moderation AI, Jun 2024. Computer vision system for automated detection of NSFW and violent content, with real-time filtering.

Vehicle Parts and Damage Detection, Mar 2024. End-to-end computer vision system using object detection and instance segmentation to identify vehicle parts, assess damage severity and generate automated assessment reports for insurance claims.

OCR Text Extraction for Identity Documents, Jun 2024. Production text extraction system for identification documents and vehicle logbooks, with a validation pipeline, deployed through FastAPI endpoints.

Credit Risk Classification, Aug 2026. Classification model in Python and scikit-learn with a multi-tab Tableau dashboard and a live risk calculator based on logistic regression.

Antimicrobial Resistance Prediction, Feb 2026. KNIME pipeline and predictive model for antibiotic resistance on a 283,715-patient Stanford Healthcare dataset, achieving an AUC of 0.784.

Business Intelligence System for a Meat Products Distributor, Aug 2026. Star schema, ETL from Access to SQL Server, two SSAS cubes and an Excel dashboard for a family-run distribution business.

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, JavaScript
- **Machine learning and AI:** TensorFlow, Keras, PyTorch, scikit-learn, YOLOv8, Mask R-CNN, OpenCV, PaddleOCR, Hugging Face, LangChain, FAISS, RAG systems
- **Big data and cloud:** PySpark, Databricks, AWS (EC2, S3, SageMaker), Azure, DuckDB, Docker
- **Analytics and BI:** Power BI, Tableau, SQL Server and SSAS, KNIME, Orange, RapidMiner, Advanced Excel
- **Engineering and methods:** Prompt engineering, FastAPI, Streamlit, Git/GitHub, CRISP-DM, Agile and Scrum

CERTIFICATIONS

- Microsoft Office Specialist (MOS), Excel, 2026
- Scrum Master Certified, Scrum Institute, 2022

LANGUAGES

English, Swahili and Meru (fluent)